Midterm (100%)

Please remember to write down your name and student number.
The procedures leading to your answers are required to grade.

1. (5%) Which of the following cannot be used as an energy source for image sensing and acquisition?
   (1). gamma rays (2) radio waves (3) acoustic (4) electronic

2. (20%) Give two points p (3, 5) and q (6, 1)
   (a) (10%) What is the chessboard distance of p and q?
   (b) (10%) What is the Euclidean distance of p and q?

3. (50%) Consider the following intensity matrix of a 5x5 image A. The depth of a pixel is 3 bits.

Image A

6	5	4	6	5
5	4	3	5	7
4	3	2	4	6
3	2	1	3	5
2	1	0	2	4

Laplacian kernel

0	-1	0
-1	4	-1
0	-1	0

Image B

0	1	6	4	6
6	6	1	5	1
4	3	3	4	6
7	7	3	3	5
2	0	1	0	2

   (a) (15%) For image A, what is the intensity matrix of the histogram-equalized image?
   (b) (15%) For image A, what is the intensity matrix of the histogram-matched image such that the output image has a histogram of the shape similar to what image B has?
   (c) (10%) What is the intensity of the pixel in the middle position (shaded one in the figure A) of the Laplacian-sharpened image, using the Laplacian kernel shown above?
   (d) (10%) What is the intensity of the pixel in the middle position (shaded one in the figure A) of the sharpened image using highboost filtering? The blurred image is obtained using a 3x3 box kernel, and k=2 for highboost filtering.

4. (15%) Suppose that we have two classes c_1 and c_2, with two patterns in the training set $x=[3\ 5\ 1]^T \in c_1$ and $x=[1\ 2\ 1]^T \in c_2$. Using the perceptron training algorithm to find the weight vector after two epoches. Note that the initial weight vector=$[0\ 0\ 0]^T$ and the learning rate is 2. Also, c_1 is on the positive side of the line (decision boundary) and c_2 is on the negative side.

5. (10%) When using bilinear interpolation, how many pixels are involved to interpolate one pixel, and why?

<END>

1. 4

0

2. (a) $\max\{|3-6|, |5-1|\} = 4$ # 10

(b) $\sqrt{(3-6)^2+(5-1)^2} = 5$ # 10

$3^2=9$ $4^2=16$

3. (a)

i	n_i	p_i
0	1	$\frac{1}{25}$
1	2	$\frac{2}{25}$
2	4	$\frac{4}{25}$
3	4	$\frac{4}{25}$
4	5	$\frac{5}{25}$
5	5	$\frac{5}{25}$
6	3	$\frac{3}{25}$
7	1	$\frac{1}{25}$

$S_0 = 7 \times \frac{1}{25} \approx 0$

$S_1 = 7 \times (\frac{1}{25} + \frac{2}{25}) = \frac{21}{25} \approx 1$

$S_2 = 7 \times (\frac{1}{25} + \frac{2}{25} + \frac{4}{25}) = \frac{49}{25} \approx 2$

$S_3 = 7 \times (\frac{1}{25} + \frac{2}{25} + \frac{4}{25} + \frac{4}{25}) = \frac{77}{25} \approx 3$

$S_4 = 7 \times (\frac{1}{25} + \frac{2}{25} + \frac{4}{25} + \frac{4}{25} + \frac{5}{25}) = \frac{112}{25} \approx 4$

$S_5 = 7 \times (\frac{1}{25} + \frac{2}{25} + \frac{4}{25} + \frac{4}{25} + \frac{5}{25} + \frac{5}{25}) = \frac{147}{25} \approx 6$

$S_6 = 7 \times (\frac{1}{25} + \frac{2}{25} + \frac{4}{25} + \frac{4}{25} + \frac{5}{25} + \frac{5}{25} + \frac{3}{25}) = \frac{168}{25} \approx 7$

$S_7 = 7 \times 1 = 7$

7	6	4	7	6
6	4	3	6	7
4	3	2	4	7
3	2	1	3	6
2	1	0	2	4

#

(b) Image B

i	n_i
0	3
1	4
2	1
3	4
4	3
5	2
6	6
7	2

$$S = T(r) = (L-1)\sum_{i=0}^{L-1} p_i$$

$$G(z) = (L-1)\sum_{i=0}^{L-1} p_z(i) = S$$

$$z = G^{-1}(S)$$

$$G_0 = 7 \cdot \frac{3}{25} \approx 1$$

$$G_1 = 7 \cdot \left(\frac{3}{25} + \frac{4}{25}\right) \approx 2$$

$$G_2 = 7 \cdot \left(\frac{3}{25} + \frac{4}{25} + \frac{1}{25}\right) \approx 2$$

$$G_3 = 7 \left(\frac{3}{25} + \frac{4}{25} + \frac{1}{25} + \frac{4}{25}\right) \approx 3$$

$$G_4 = 7 \left(\frac{3}{25} + \frac{4}{25} + \frac{1}{25} + \frac{4}{25} + \frac{3}{25}\right) \approx 4$$

$$G_5 = 7 \left(\frac{3}{25} + \frac{4}{25} + \frac{1}{25} + \frac{4}{25} + \frac{3}{25} + \frac{2}{25}\right) \approx 5$$

$$G_6 = 7 \left(\frac{3}{25} + \frac{4}{25} + \frac{1}{25} + \frac{4}{25} + \frac{3}{25} + \frac{2}{25} + \frac{6}{25}\right) \approx 6$$

$$G_7 = 7$$

$$z = G^{-1}(S) \Rightarrow$$

S:

7	6	4	7	6
6	4	3	6	7
4	3	2	4	7
3	2	1	3	6
2	1	0	2	4

7	6	4	7	6
6	4	3	6	7
4	3	1	4	7
3	1	0	3	6
1	0	0	1	4

#

(c) g(

+9.

(d)

+7.

4. epo

epoc

(c) $g(x,y) = f(x,y) + \Delta^2 f$

+9.

$= 2 + (-3 - 3 - 4 - 1 + 8)$

$= \cancel{+1}$

$\rightarrow 0$ #

(d) $g(x,y) = f(x,y) + K \cdot \underbrace{mask(x,y)}_{} = f(x,y) - \bar{f}(x,y)$

+7.

$= 2 + 2\left(\frac{4+3+5+3+2+4+2+1+3}{9}\right)$ 27

$= 2 + 6 = 8$ #

4. epoch 1:

for x1: $[0\ 0\ 0]\begin{bmatrix}3\\5\\1\end{bmatrix} = 0 \leq 0$

$W_1^T = [000] + 2[3\ 5\ 1] = [6\ 10\ 2]$

for x2: $[6\ 10\ 2]\begin{bmatrix}1\\2\\1\end{bmatrix} = 6 + 20 + 2 = 28 > 0$

$W_2^T = [6\ 10\ 2] - 2[1\ 2\ 1] = [4\ 6\ 0]$

epoch 2:

for x1: $[4\ 6\ 0]\begin{bmatrix}3\\5\\1\end{bmatrix} = 12 + 30 = 42 > 0$

for x2: $[4\ 6\ 0]\begin{bmatrix}1\\2\\1\end{bmatrix} = 4 + 12 = 16 > 0$

$W_3^T = [4\ 6\ 0] - 2[1\ 2\ 1] = [2\ 2\ -2]$

weight vector $= [2\ 2\ -2]^T$

#

5. $g(x,y) = \underline{ax+by+cxy+d}_{4\text{pixel}}$

4 pixels are involved #